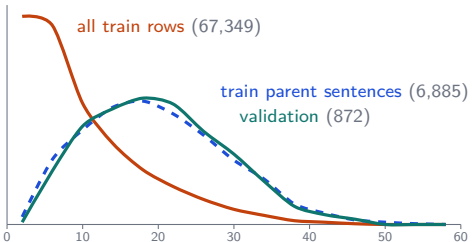
1 · Data and metrics

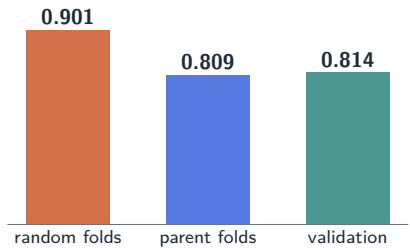
1 Train rows are mostly short sub-phrases

share of rows by length, in tokens (SST-2 keeps every parse-tree node)



2 Random folds leak, parent folds do not

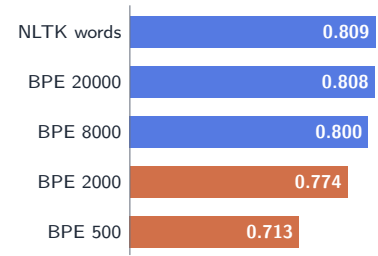
accuracy of log-linear (NLTK, token indicators)



2 · Features

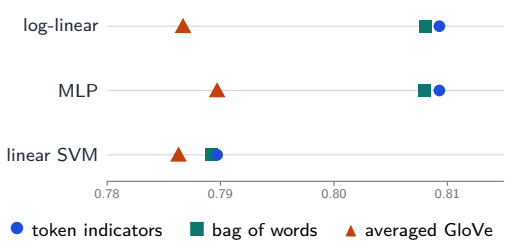
1 Large vocabularies win

log-linear, token indicators: CV accuracy by tokeniser



2 Averaged GloVe is worse for every model

CV accuracy (3 seeds × 5 folds) by representation

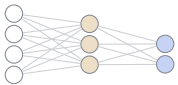


3 · Architectures

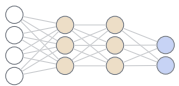
1 Four architectures

two from the tutorial, two of our own

log-linear
logistic regression



MLP
1 hidden layer, 32 units

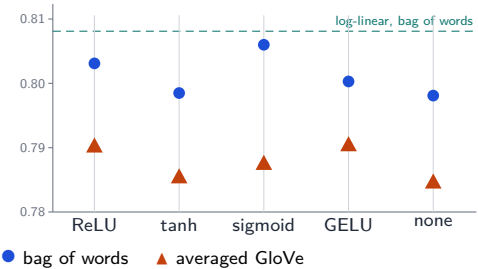


2-layer MLP (ours)
2 hidden layers, 128 units

linear SVM (ours)
squared hinge loss

2 The non-linearity barely matters

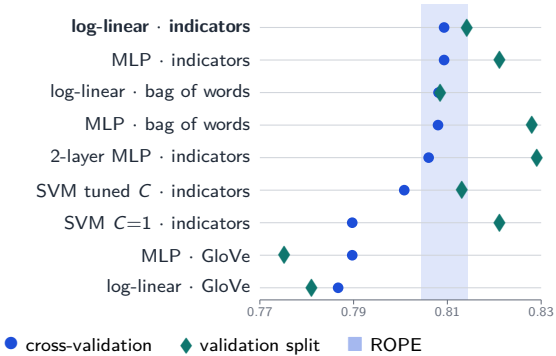
MLP: CV accuracy by activation function



4 · Results

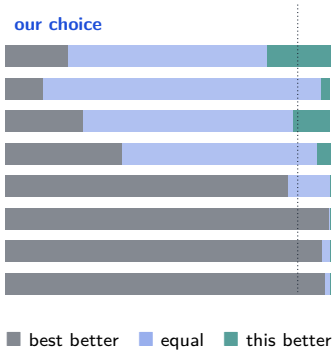
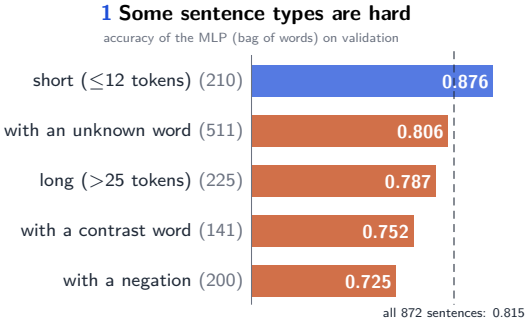
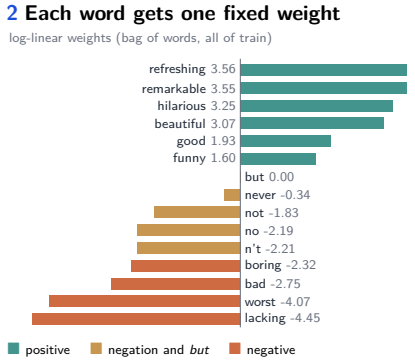
1 Log-linear ties for best

accuracy on NLTK tokens



2 Most differences are undecided

posterior vs the best system, ROPE 0.005



5 · Extra: why every model stops at about 81%